Ogbonna Arinze

arinzeogbo@gmail.com | LinkedIn

SUMMARY

Senior Full-Stack Web Engineer with 8 years of professional experience building and scaling web platforms across healthcare, fintech, civic tech, and enterprise environments. Deep expertise in **React, Next.js, TypeScript**, and **Tailwind CSS** on the frontend, paired with strong backend capabilities in **NodeJS, NestJS, REST API design**, and **PostgreSQL**. Experienced in owning features end-to-end from database schema design through CI/CD deployment, with production experience in **Docker**, automated testing (Playwright, Jest), and performance optimization. Passionate about building reliable, well-tested systems and mentoring engineering teams.

CORE COMPETENCIES

- **Frontend:** React, Next.js, TypeScript, Tailwind CSS, Vue.js, Redux, Zustand, Radix UI, SSR/SSG, and responsive design.
- **Backend & APIs:** NodeJS, NestJS, RESTful API design, GraphQL, PostgreSQL, MongoDB, Prisma ORM, JWT authentication, and RBAC.
- **Infrastructure & DevOps:** Docker, Docker Compose, GitHub Actions, CI/CD pipelines, AWS (CloudFront, S3), Vercel, Traefik, and Nx monorepo orchestration.
- **Testing & Quality:** Playwright (E2E), Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Performance & SEO:** Core Web Vitals, Lighthouse, code splitting, CDN strategies, bundle analysis, Open Graph, JSON-LD, and SSR-based SEO.
- **AI & Data:** RAG pipelines, OpenAI API, LangChain, vector search (pgvector, Chroma), Recharts, and react-chartjs data visualization.
- **Leadership:** Technical architecture reviews, mentorship, Agile/Scrum, TDD/BDD, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) San Francisco, CA (Remote)

Frontend Engineer *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React** Chrome extension, refactoring legacy components while shipping features in parallel during the migration period.
- Co-developed a full-stack **Next.js** chat widget powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to connect data sources, embed the widget on external sites, and query their data in real time.
- Architected a reusable **design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Implemented **JWT-based authentication** and **role-based access control** on the frontend using HTTP-only cookies for secure token handling, collaborating with backend engineers to align on auth strategies and permission-based UI rendering.
- Optimized application performance by implementing **lazy loading, code splitting**, and **CDN asset delivery**, identifying and resolving bottlenecks with **Lighthouse** and Chrome DevTools to significantly reduce load times.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline** via **GitHub Actions**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio Lagos, Nigeria (Hybrid)

Senior Frontend Developer *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** automated testing frameworks, leading to a **20% decrease** in bug reports and improved release confidence.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices including user story estimation and sprint planning.
- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes across the team.

- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, handling thousands to millions of daily transactions and boosting settlement rates by **80%**, developing critical modules for MasterCard, VISA, Express-cash, and E-commerce using **React, Redux, RxJS**, and **Axios**.
- Developed a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design**.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, focusing on testable, maintainable, and well-documented code.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** (testing) and **AWS** (production).
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with **React 19** and **Tailwind CSS v4** (22 interactive chart types via **Recharts**), Next.js admin dashboard, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and built **RESTful APIs** with NestJS powering all client applications.
- Configured **CI/CD** with **GitHub Actions** for automated builds, **Playwright** E2E tests (4 test projects), and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, memory limits (API 1GB, Ingest 4GB, Postgres 2GB), and 30s graceful shutdown. Integrated **Traefik** for reverse proxy routing.
- Implemented **SSE streaming** for real-time chat responses with keepalive heartbeats, phone OTP authentication, **Telegram bot** with webhook handling, and shareable public conversations with **SEO optimization** (Open Graph metadata, JSON-LD FAQPage schema, slug-based URLs with SSR).
- Built a document ingestion pipeline supporting PDF (with OCR), XLSX/XLS, DOCX, JSON, and Markdown, processing **700+ documents** (959 files) across 37 Nigerian states with zero ingestion errors.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*